R Notebook

```
library(dada2); packageVersion("dada2")
```

```
## Loading required package: Rcpp
```

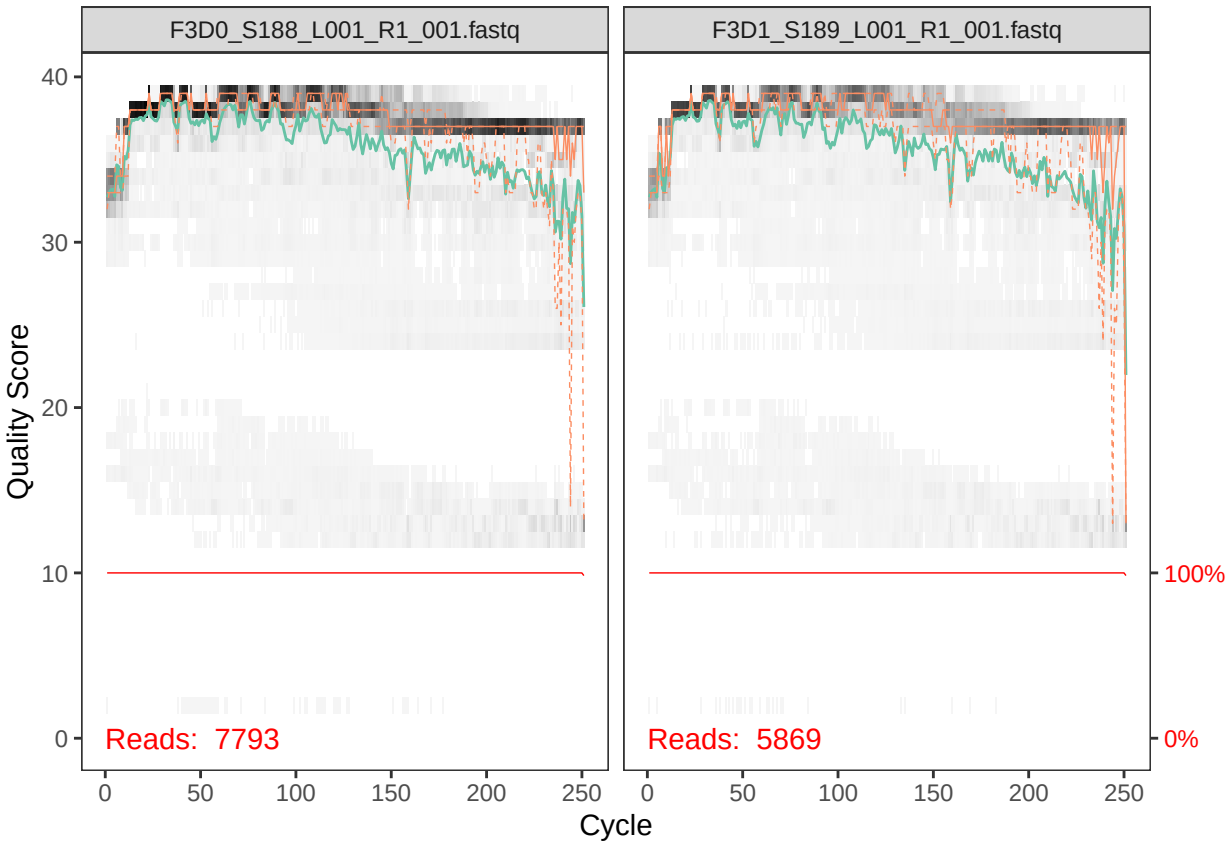
```
## [1] '1.40.0'
```

```
path <- "/home/rstudio/Dada2/Dada2/MiSeq_SOP" # CHANGE ME to the directory containing the fastq files a
list.files(path)
```

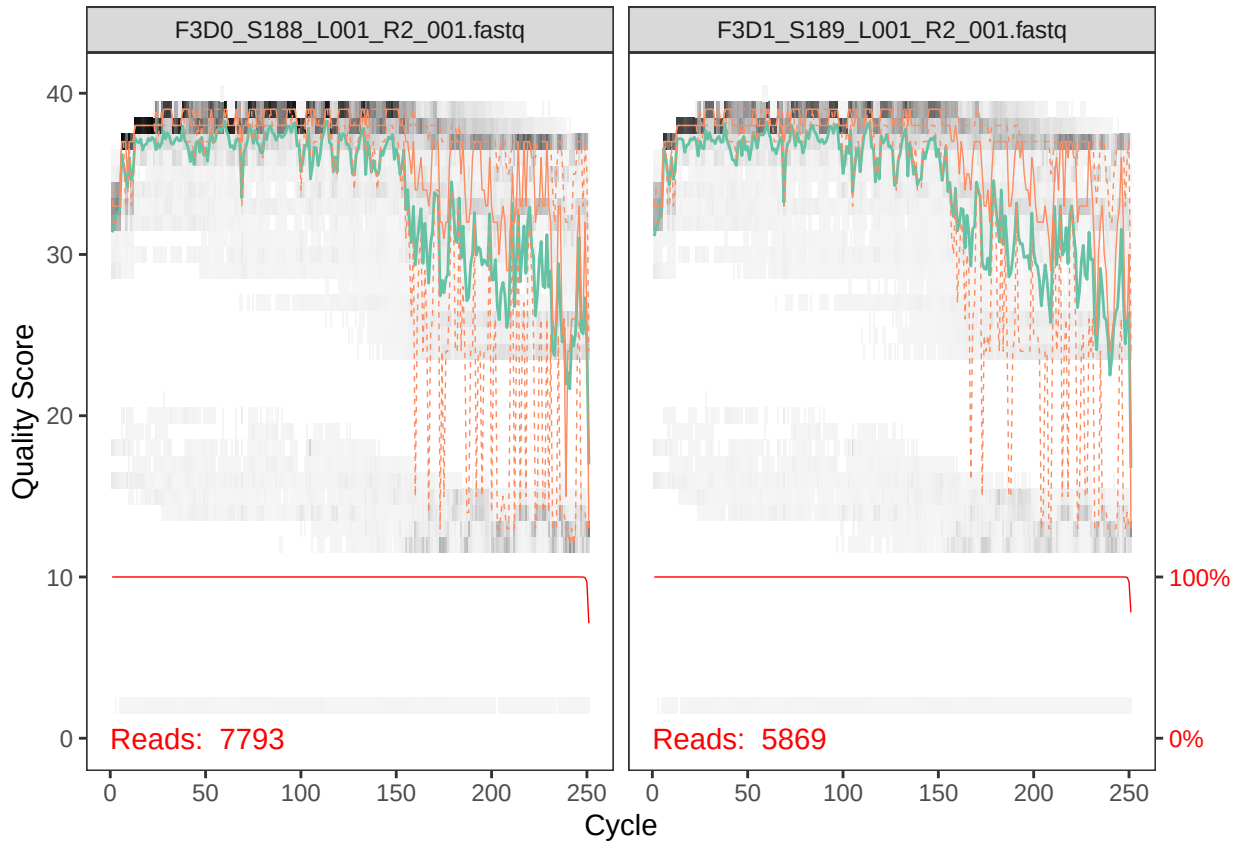
```
##  [1] "F3D0_S188_L001_R1_001.fastq"   "F3D0_S188_L001_R2_001.fastq"
##  [3] "F3D1_S189_L001_R1_001.fastq"   "F3D1_S189_L001_R2_001.fastq"
##  [5] "F3D141_S207_L001_R1_001.fastq" "F3D141_S207_L001_R2_001.fastq"
##  [7] "F3D142_S208_L001_R1_001.fastq" "F3D142_S208_L001_R2_001.fastq"
##  [9] "F3D143_S209_L001_R1_001.fastq" "F3D143_S209_L001_R2_001.fastq"
## [11] "F3D144_S210_L001_R1_001.fastq" "F3D144_S210_L001_R2_001.fastq"
## [13] "F3D145_S211_L001_R1_001.fastq" "F3D145_S211_L001_R2_001.fastq"
## [15] "F3D146_S212_L001_R1_001.fastq" "F3D146_S212_L001_R2_001.fastq"
## [17] "F3D147_S213_L001_R1_001.fastq" "F3D147_S213_L001_R2_001.fastq"
## [19] "F3D148_S214_L001_R1_001.fastq" "F3D148_S214_L001_R2_001.fastq"
## [21] "F3D149_S215_L001_R1_001.fastq" "F3D149_S215_L001_R2_001.fastq"
## [23] "F3D150_S216_L001_R1_001.fastq" "F3D150_S216_L001_R2_001.fastq"
## [25] "F3D2_S190_L001_R1_001.fastq"   "F3D2_S190_L001_R2_001.fastq"
## [27] "F3D3_S191_L001_R1_001.fastq"   "F3D3_S191_L001_R2_001.fastq"
## [29] "F3D5_S193_L001_R1_001.fastq"   "F3D5_S193_L001_R2_001.fastq"
## [31] "F3D6_S194_L001_R1_001.fastq"   "F3D6_S194_L001_R2_001.fastq"
## [33] "F3D7_S195_L001_R1_001.fastq"   "F3D7_S195_L001_R2_001.fastq"
## [35] "F3D8_S196_L001_R1_001.fastq"   "F3D8_S196_L001_R2_001.fastq"
## [37] "F3D9_S197_L001_R1_001.fastq"   "F3D9_S197_L001_R2_001.fastq"
## [39] "filtered"                      "HMP_MOCK.v35.fasta"
## [41] "Mock_S280_L001_R1_001.fastq"   "Mock_S280_L001_R2_001.fastq"
## [43] "mouse.dpw.metadata"            "mouse.time.design"
## [45] "stability.batch"               "stability.files"
```

```
# Forward and reverse fastq filenames have format: SAMPLENAME_R1_001.fastq and SAMPLENAME_R2_001.fastq
fnFs <- sort(list.files(path, pattern="_R1_001.fastq", full.names = TRUE))
fnRs <- sort(list.files(path, pattern="_R2_001.fastq", full.names = TRUE))
# Extract sample names, assuming filenames have format: SAMPLENAME_XXX.fastq
sample.names <- sapply(strsplit(basename(fnFs), "_"), `[`, 1)
```

```
plotQualityProfile(fnFs[1:2])
```




```
plotQualityProfile(fnRs[1:2])
```




```
# Place filtered files in filtered/ subdirectory
filtFs <- file.path(path, "filtered", paste0(sample.names, "_F_filt.fastq.gz"))
filtRs <- file.path(path, "filtered", paste0(sample.names, "_R_filt.fastq.gz"))
names(filtFs) <- sample.names
names(filtRs) <- sample.names
```

```
out <- filterAndTrim(fnFs, filtFs, fnRs, filtRs, truncLen=c(240,160),
              maxN=0, maxEE=c(2,2), truncQ=2, rm.phix=TRUE,
              compress=TRUE, multithread=FALSE) # On Windows set multithread=FALSE (only needed for fil
head(out)
```

```
##                                reads.in reads.out
## F3D0_S188_L001_R1_001.fastq        7793      7113
## F3D1_S189_L001_R1_001.fastq        5869      5299
## F3D141_S207_L001_R1_001.fastq      5958      5463
## F3D142_S208_L001_R1_001.fastq      3183      2914
## F3D143_S209_L001_R1_001.fastq      3178      2941
## F3D144_S210_L001_R1_001.fastq      4827      4312
```

```
errF <- learnErrors(filtFs, multithread=TRUE)
```

```
## 33514080 total bases in 139642 reads from 20 samples will be used for learning the error rates.
```

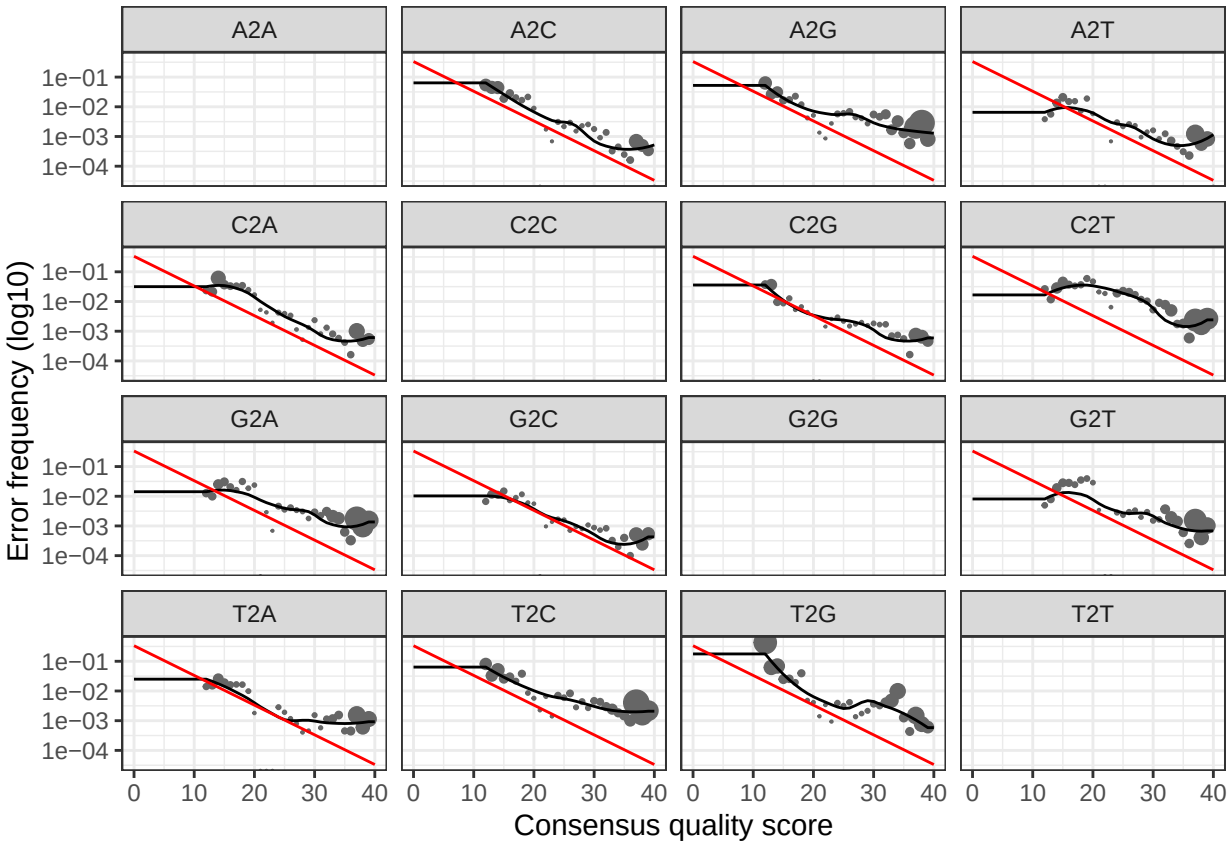
```
errR <- learnErrors(filtRs, multithread=TRUE)
```

```
## 22342720 total bases in 139642 reads from 20 samples will be used for learning the error rates.
```

```
plotErrors(errF, nominalQ=TRUE)
```

```
## Warning in scale_y_log10(): log-10 transformation introduced infinite values.
```

```
## Warning: Removed 164 rows containing missing values or values outside the scale range
## (`geom_line()`).
## Removed 164 rows containing missing values or values outside the scale range
## (`geom_line()`).
```




```
dadaFs <- dada(filtFs, err=errF, multithread=TRUE)
```

```
## Sample 1 - 7113 reads in 1979 unique sequences.
## Sample 2 - 5299 reads in 1639 unique sequences.
## Sample 3 - 5463 reads in 1477 unique sequences.
## Sample 4 - 2914 reads in 904 unique sequences.
## Sample 5 - 2941 reads in 939 unique sequences.
## Sample 6 - 4312 reads in 1267 unique sequences.
## Sample 7 - 6741 reads in 1756 unique sequences.
## Sample 8 - 4560 reads in 1438 unique sequences.
```

```
## Sample 9 - 15637 reads in 3590 unique sequences.
## Sample 10 - 11413 reads in 2762 unique sequences.
## Sample 11 - 12017 reads in 3021 unique sequences.
## Sample 12 - 5032 reads in 1566 unique sequences.
## Sample 13 - 18075 reads in 3707 unique sequences.
## Sample 14 - 6250 reads in 1479 unique sequences.
## Sample 15 - 4052 reads in 1195 unique sequences.
## Sample 16 - 7369 reads in 1832 unique sequences.
## Sample 17 - 4765 reads in 1183 unique sequences.
## Sample 18 - 4871 reads in 1382 unique sequences.
## Sample 19 - 6504 reads in 1709 unique sequences.
## Sample 20 - 4314 reads in 897 unique sequences.
```

```
dadaRs <- dada(filtRs, err=errR, multithread=TRUE)
```

```
## Sample 1 - 7113 reads in 1660 unique sequences.
## Sample 2 - 5299 reads in 1349 unique sequences.
## Sample 3 - 5463 reads in 1335 unique sequences.
## Sample 4 - 2914 reads in 853 unique sequences.
## Sample 5 - 2941 reads in 880 unique sequences.
## Sample 6 - 4312 reads in 1286 unique sequences.
## Sample 7 - 6741 reads in 1803 unique sequences.
## Sample 8 - 4560 reads in 1265 unique sequences.
## Sample 9 - 15637 reads in 3414 unique sequences.
## Sample 10 - 11413 reads in 2522 unique sequences.
## Sample 11 - 12017 reads in 2771 unique sequences.
## Sample 12 - 5032 reads in 1415 unique sequences.
## Sample 13 - 18075 reads in 3290 unique sequences.
## Sample 14 - 6250 reads in 1390 unique sequences.
## Sample 15 - 4052 reads in 1134 unique sequences.
## Sample 16 - 7369 reads in 1635 unique sequences.
## Sample 17 - 4765 reads in 1084 unique sequences.
## Sample 18 - 4871 reads in 1161 unique sequences.
## Sample 19 - 6504 reads in 1502 unique sequences.
## Sample 20 - 4314 reads in 732 unique sequences.
```

```
dadaFs[[1]]
```

```
## dada-class: object describing DADA2 denoising results
## 128 sequence variants were inferred from 1979 input unique sequences.
## Key parameters: OMEGA_A = 1e-40, OMEGA_C = 1e-40, BAND_SIZE = 16
```

```
mergers <- mergePairs(dadaFs, filtFs, dadaRs, filtRs, verbose=TRUE)
```

```
## 6540 paired-reads (in 107 unique pairings) successfully merged out of 6891 (in 197 pairings) input.

## 5028 paired-reads (in 101 unique pairings) successfully merged out of 5190 (in 157 pairings) input.

## 4986 paired-reads (in 81 unique pairings) successfully merged out of 5267 (in 166 pairings) input.

## 2595 paired-reads (in 52 unique pairings) successfully merged out of 2754 (in 108 pairings) input.
```

```
## 2553 paired-reads (in 60 unique pairings) successfully merged out of 2785 (in 119 pairings) input.

## 3646 paired-reads (in 55 unique pairings) successfully merged out of 4109 (in 157 pairings) input.

## 6079 paired-reads (in 81 unique pairings) successfully merged out of 6514 (in 198 pairings) input.

## 3968 paired-reads (in 91 unique pairings) successfully merged out of 4388 (in 187 pairings) input.

## 14233 paired-reads (in 143 unique pairings) successfully merged out of 15355 (in 352 pairings) input

## 10528 paired-reads (in 120 unique pairings) successfully merged out of 11165 (in 278 pairings) input

## 11154 paired-reads (in 137 unique pairings) successfully merged out of 11797 (in 298 pairings) input

## 4349 paired-reads (in 85 unique pairings) successfully merged out of 4802 (in 179 pairings) input.

## 17431 paired-reads (in 153 unique pairings) successfully merged out of 17812 (in 272 pairings) input

## 5850 paired-reads (in 81 unique pairings) successfully merged out of 6095 (in 159 pairings) input.

## 3716 paired-reads (in 86 unique pairings) successfully merged out of 3894 (in 147 pairings) input.

## 6865 paired-reads (in 99 unique pairings) successfully merged out of 7191 (in 187 pairings) input.

## 4426 paired-reads (in 67 unique pairings) successfully merged out of 4603 (in 127 pairings) input.

## 4576 paired-reads (in 101 unique pairings) successfully merged out of 4739 (in 174 pairings) input.

## 6092 paired-reads (in 109 unique pairings) successfully merged out of 6315 (in 173 pairings) input.

## 4269 paired-reads (in 20 unique pairings) successfully merged out of 4281 (in 28 pairings) input.
```

```
# Inspect the merger data.frame from the first sample
head(mergers[[1]])
```

```
##
## 1 TACGGAGGATGCGAGCGTTATCCGGATTTATTGGGTTTAAAGGGTGCGCAGGCGGAAGATCAAGTCAGCGGTAAAATTGAGAGGCTCAACCTCTTCGAG
## 2 TACGGAGGATGCGAGCGTTATCCGGATTTATTGGGTTTAAAGGGTGCGTAGGCGGCCTGCCAAGTCAGCGGTAAAATTGCGGGGCTCAACCCCGTACAG
## 3 TACGGAGGATGCGAGCGTTATCCGGATTTATTGGGTTTAAAGGGTGCGTAGGCGGGCTGTTAAGTCAGCGGTCAAATGTCGGGGCTCAACCCCGGCCTG
## 4 TACGGAGGATGCGAGCGTTATCCGGATTTATTGGGTTTAAAGGGTGCGTAGGCGGGCTTTTAAGTCAGCGGTAAAAATTCGGGGCTCAACCCCGTCCGG
## 5 TACGGAGGATGCGAGCGTTATCCGGATTTATTGGGTTTAAAGGGTGCGCAGGCGGACTCTCAAGTCAGCGGTCAAATCGCGGGGCTCAACCCCGTTCCG
## 6 TACGGAGGATGCGAGCGTTATCCGGATTTATTGGGTTTAAAGGGTGCGTAGGCGGGATGCCAAGTCAGCGGTAAAAAAGCGGTGCTCAACGCCGTCGAG
##   abundance forward reverse nmatch nmismatch nindel prefer accept
## 1       579       1       1    148         0      0      1   TRUE
## 2       470       2       2    148         0      0      2   TRUE
## 3       449       3       4    148         0      0      1   TRUE
## 4       430       4       3    148         0      0      2   TRUE
## 5       345       5       6    148         0      0      1   TRUE
## 6       282       6       5    148         0      0      2   TRUE
```

```r
seqtab <- makeSequenceTable(mergers)
dim(seqtab)
```

```
## [1]  20 293
```

```r
# Inspect distribution of sequence lengths
table(nchar(getSequences(seqtab)))
```

```
##
## 251 252 253 254 255
##   1  88 196   6   2
```

```r
seqtab.nochim <- removeBimeraDenovo(seqtab, method="consensus", multithread=TRUE, verbose=TRUE)
```

```
## Identified 61 bimeras out of 293 input sequences.
```

```r
dim(seqtab.nochim)
```

```
## [1]  20 232
```

```r
sum(seqtab.nochim)/sum(seqtab)
```

```
## [1] 0.9640374
```

```r
getN <- function(x) sum(getUniques(x))
track <- cbind(out, sapply(dadaFs, getN), sapply(dadaRs, getN), sapply(mergers, getN), rowSums(seqtab.no
# If processing a single sample, remove the sapply calls: e.g. replace sapply(dadaFs, getN) with getN(d
colnames(track) <- c("input", "filtered", "denoisedF", "denoisedR", "merged", "nonchim")
rownames(track) <- sample.names
head(track)
```

```
##        input filtered denoisedF denoisedR merged nonchim
## F3D0    7793     7113      6976      6979   6540    6528
## F3D1    5869     5299      5227      5239   5028    5017
## F3D141  5958     5463      5331      5357   4986    4863
## F3D142  3183     2914      2799      2830   2595    2521
## F3D143  3178     2941      2822      2868   2553    2519
## F3D144  4827     4312      4151      4228   3646    3507
```

```r
library(dada2)

# Lancer l'assignation taxonomique avec le bon chemin
taxa <- assignTaxonomy(seqtab.nochim, "silva_nr99_v138.2_toGenus_trainset.fa.gz", multithread = TRUE)

# Vérifier les résultats
head(taxa)
```

```
##
## TACGGAGGATGCGAGCGTTATCCGGATTTATTGGGTTTAAAGGGTGCGCAGGCGGAAGATCAAGTCAGCGGTAAAATTGAGAGGCTCAACCTCTTCGAGC
## TACGGAGGATGCGAGCGTTATCCGGATTTATTGGGTTTAAAGGGTGCGCAGGCGGACTCTCAAGTCAGCGGTCAAATCGCGGGGCTCAACCCCGTTCCGC
## TACGGAGGATGCGAGCGTTATCCGGATTTATTGGGTTTAAAGGGTGCGTAGGCGGGCTGTTAAGTCAGCGGTCAAATGTCGGGGCTCAACCCCGGCCTGC
## TACGGAGGATGCGAGCGTTATCCGGATTTATTGGGTTTAAAGGGTGCGTAGGCGGGCTTTTAAGTCAGCGGTAAAAATTCGGGGCTCAACCCCGTCCGGC
## TACGGAGGATCCGAGCGTTATCCGGATTTATTGGGTTTAAAGGGAGCGTAGGTGGATTGTTAAGTCAGTTGTGAAAGTTTGCGGCTCAACCGTAAAATTG
## TACGGAGGATGCGAGCGTTATCCGGATTTATTGGGTTTAAAGGGTGCGTAGGCGGCCTGCCAAGTCAGCGGTAAAATTGCGGGGCTCAACCCCGTACAGC
##
## TACGGAGGATGCGAGCGTTATCCGGATTTATTGGGTTTAAAGGGTGCGCAGGCGGAAGATCAAGTCAGCGGTAAAATTGAGAGGCTCAACCTCTTCGAGC
## TACGGAGGATGCGAGCGTTATCCGGATTTATTGGGTTTAAAGGGTGCGCAGGCGGACTCTCAAGTCAGCGGTCAAATCGCGGGGCTCAACCCCGTTCCGC
## TACGGAGGATGCGAGCGTTATCCGGATTTATTGGGTTTAAAGGGTGCGTAGGCGGGCTGTTAAGTCAGCGGTCAAATGTCGGGGCTCAACCCCGGCCTGC
## TACGGAGGATGCGAGCGTTATCCGGATTTATTGGGTTTAAAGGGTGCGTAGGCGGGCTTTTAAGTCAGCGGTAAAAATTCGGGGCTCAACCCCGTCCGGC
## TACGGAGGATCCGAGCGTTATCCGGATTTATTGGGTTTAAAGGGAGCGTAGGTGGATTGTTAAGTCAGTTGTGAAAGTTTGCGGCTCAACCGTAAAATTG
## TACGGAGGATGCGAGCGTTATCCGGATTTATTGGGTTTAAAGGGTGCGTAGGCGGCCTGCCAAGTCAGCGGTAAAATTGCGGGGCTCAACCCCGTACAGC
##
## TACGGAGGATGCGAGCGTTATCCGGATTTATTGGGTTTAAAGGGTGCGCAGGCGGAAGATCAAGTCAGCGGTAAAATTGAGAGGCTCAACCTCTTCGAGC
## TACGGAGGATGCGAGCGTTATCCGGATTTATTGGGTTTAAAGGGTGCGCAGGCGGACTCTCAAGTCAGCGGTCAAATCGCGGGGCTCAACCCCGTTCCGC
## TACGGAGGATGCGAGCGTTATCCGGATTTATTGGGTTTAAAGGGTGCGTAGGCGGGCTGTTAAGTCAGCGGTCAAATGTCGGGGCTCAACCCCGGCCTGC
## TACGGAGGATGCGAGCGTTATCCGGATTTATTGGGTTTAAAGGGTGCGTAGGCGGGCTTTTAAGTCAGCGGTAAAAATTCGGGGCTCAACCCCGTCCGGC
## TACGGAGGATCCGAGCGTTATCCGGATTTATTGGGTTTAAAGGGAGCGTAGGTGGATTGTTAAGTCAGTTGTGAAAGTTTGCGGCTCAACCGTAAAATTG
## TACGGAGGATGCGAGCGTTATCCGGATTTATTGGGTTTAAAGGGTGCGTAGGCGGCCTGCCAAGTCAGCGGTAAAATTGCGGGGCTCAACCCCGTACAGC
##
## TACGGAGGATGCGAGCGTTATCCGGATTTATTGGGTTTAAAGGGTGCGCAGGCGGAAGATCAAGTCAGCGGTAAAATTGAGAGGCTCAACCTCTTCGAGC
## TACGGAGGATGCGAGCGTTATCCGGATTTATTGGGTTTAAAGGGTGCGCAGGCGGACTCTCAAGTCAGCGGTCAAATCGCGGGGCTCAACCCCGTTCCGC
## TACGGAGGATGCGAGCGTTATCCGGATTTATTGGGTTTAAAGGGTGCGTAGGCGGGCTGTTAAGTCAGCGGTCAAATGTCGGGGCTCAACCCCGGCCTGC
## TACGGAGGATGCGAGCGTTATCCGGATTTATTGGGTTTAAAGGGTGCGTAGGCGGGCTTTTAAGTCAGCGGTAAAAATTCGGGGCTCAACCCCGTCCGGC
## TACGGAGGATCCGAGCGTTATCCGGATTTATTGGGTTTAAAGGGAGCGTAGGTGGATTGTTAAGTCAGTTGTGAAAGTTTGCGGCTCAACCGTAAAATTG
## TACGGAGGATGCGAGCGTTATCCGGATTTATTGGGTTTAAAGGGTGCGTAGGCGGCCTGCCAAGTCAGCGGTAAAATTGCGGGGCTCAACCCCGTACAGC
##
## TACGGAGGATGCGAGCGTTATCCGGATTTATTGGGTTTAAAGGGTGCGCAGGCGGAAGATCAAGTCAGCGGTAAAATTGAGAGGCTCAACCTCTTCGAGC
## TACGGAGGATGCGAGCGTTATCCGGATTTATTGGGTTTAAAGGGTGCGCAGGCGGACTCTCAAGTCAGCGGTCAAATCGCGGGGCTCAACCCCGTTCCGC
## TACGGAGGATGCGAGCGTTATCCGGATTTATTGGGTTTAAAGGGTGCGTAGGCGGGCTGTTAAGTCAGCGGTCAAATGTCGGGGCTCAACCCCGGCCTGC
## TACGGAGGATGCGAGCGTTATCCGGATTTATTGGGTTTAAAGGGTGCGTAGGCGGGCTTTTAAGTCAGCGGTAAAAATTCGGGGCTCAACCCCGTCCGGC
## TACGGAGGATCCGAGCGTTATCCGGATTTATTGGGTTTAAAGGGAGCGTAGGTGGATTGTTAAGTCAGTTGTGAAAGTTTGCGGCTCAACCGTAAAATTG
## TACGGAGGATGCGAGCGTTATCCGGATTTATTGGGTTTAAAGGGTGCGTAGGCGGCCTGCCAAGTCAGCGGTAAAATTGCGGGGCTCAACCCCGTACAGC
```

```
library(dada2)
# Remplacez le nom par le nom exact du fichier qui est apparu dans l'onglet Files
taxa2 <- addSpecies(taxa, "silva_v138.2_assignSpecies.fa.gz")
head(taxa)
```

```
##
## TACGGAGGATGCGAGCGTTATCCGGATTTATTGGGTTTAAAGGGTGCGCAGGCGGAAGATCAAGTCAGCGGTAAAATTGAGAGGCTCAACCTCTTCGAGC
## TACGGAGGATGCGAGCGTTATCCGGATTTATTGGGTTTAAAGGGTGCGCAGGCGGACTCTCAAGTCAGCGGTCAAATCGCGGGGCTCAACCCCGTTCCGC
## TACGGAGGATGCGAGCGTTATCCGGATTTATTGGGTTTAAAGGGTGCGTAGGCGGGCTGTTAAGTCAGCGGTCAAATGTCGGGGCTCAACCCCGGCCTGC
## TACGGAGGATGCGAGCGTTATCCGGATTTATTGGGTTTAAAGGGTGCGTAGGCGGGCTTTTAAGTCAGCGGTAAAAATTCGGGGCTCAACCCCGTCCGGC
## TACGGAGGATCCGAGCGTTATCCGGATTTATTGGGTTTAAAGGGAGCGTAGGTGGATTGTTAAGTCAGTTGTGAAAGTTTGCGGCTCAACCGTAAAATTG
```

```
## TACGGAGGATGCGAGCGTTATCCGGATTTATTGGGTTTAAAGGGTGCGTAGGCGGCCTGCCAAGTCAGCGGTAAAATTGCGGGGCTCAACCCCGTACAGC
##
## TACGGAGGATGCGAGCGTTATCCGGATTTATTGGGTTTAAAGGGTGCGCAGGCGGAAGATCAAGTCAGCGGTAAAATTGAGAGGCTCAACCTCTTCGAGC
## TACGGAGGATGCGAGCGTTATCCGGATTTATTGGGTTTAAAGGGTGCGCAGGCGGACTCTCAAGTCAGCGGTCAAATCGCGGGGCTCAACCCCGTTCCGC
## TACGGAGGATGCGAGCGTTATCCGGATTTATTGGGTTTAAAGGGTGCGTAGGCGGGCTGTTAAGTCAGCGGTCAAATGTCGGGGCTCAACCCCGGCCTGC
## TACGGAGGATGCGAGCGTTATCCGGATTTATTGGGTTTAAAGGGTGCGTAGGCGGGCTTTTAAGTCAGCGGTAAAAATTCGGGGCTCAACCCCGTCCGGC
## TACGGAGGATCCGAGCGTTATCCGGATTTATTGGGTTTAAAGGGAGCGTAGGTGGATTGTTAAGTCAGTTGTGAAAGTTTGCGGCTCAACCGTAAAATTG
## TACGGAGGATGCGAGCGTTATCCGGATTTATTGGGTTTAAAGGGTGCGTAGGCGGCCTGCCAAGTCAGCGGTAAAATTGCGGGGCTCAACCCCGTACAGC
##
## TACGGAGGATGCGAGCGTTATCCGGATTTATTGGGTTTAAAGGGTGCGCAGGCGGAAGATCAAGTCAGCGGTAAAATTGAGAGGCTCAACCTCTTCGAGC
## TACGGAGGATGCGAGCGTTATCCGGATTTATTGGGTTTAAAGGGTGCGCAGGCGGACTCTCAAGTCAGCGGTCAAATCGCGGGGCTCAACCCCGTTCCGC
## TACGGAGGATGCGAGCGTTATCCGGATTTATTGGGTTTAAAGGGTGCGTAGGCGGGCTGTTAAGTCAGCGGTCAAATGTCGGGGCTCAACCCCGGCCTGC
## TACGGAGGATGCGAGCGTTATCCGGATTTATTGGGTTTAAAGGGTGCGTAGGCGGGCTTTTAAGTCAGCGGTAAAAATTCGGGGCTCAACCCCGTCCGGC
## TACGGAGGATCCGAGCGTTATCCGGATTTATTGGGTTTAAAGGGAGCGTAGGTGGATTGTTAAGTCAGTTGTGAAAGTTTGCGGCTCAACCGTAAAATTG
## TACGGAGGATGCGAGCGTTATCCGGATTTATTGGGTTTAAAGGGTGCGTAGGCGGCCTGCCAAGTCAGCGGTAAAATTGCGGGGCTCAACCCCGTACAGC
##
## TACGGAGGATGCGAGCGTTATCCGGATTTATTGGGTTTAAAGGGTGCGCAGGCGGAAGATCAAGTCAGCGGTAAAATTGAGAGGCTCAACCTCTTCGAGC
## TACGGAGGATGCGAGCGTTATCCGGATTTATTGGGTTTAAAGGGTGCGCAGGCGGACTCTCAAGTCAGCGGTCAAATCGCGGGGCTCAACCCCGTTCCGC
## TACGGAGGATGCGAGCGTTATCCGGATTTATTGGGTTTAAAGGGTGCGTAGGCGGGCTGTTAAGTCAGCGGTCAAATGTCGGGGCTCAACCCCGGCCTGC
## TACGGAGGATGCGAGCGTTATCCGGATTTATTGGGTTTAAAGGGTGCGTAGGCGGGCTTTTAAGTCAGCGGTAAAAATTCGGGGCTCAACCCCGTCCGGC
## TACGGAGGATCCGAGCGTTATCCGGATTTATTGGGTTTAAAGGGAGCGTAGGTGGATTGTTAAGTCAGTTGTGAAAGTTTGCGGCTCAACCGTAAAATTG
## TACGGAGGATGCGAGCGTTATCCGGATTTATTGGGTTTAAAGGGTGCGTAGGCGGCCTGCCAAGTCAGCGGTAAAATTGCGGGGCTCAACCCCGTACAGC
##
## TACGGAGGATGCGAGCGTTATCCGGATTTATTGGGTTTAAAGGGTGCGCAGGCGGAAGATCAAGTCAGCGGTAAAATTGAGAGGCTCAACCTCTTCGAGC
## TACGGAGGATGCGAGCGTTATCCGGATTTATTGGGTTTAAAGGGTGCGCAGGCGGACTCTCAAGTCAGCGGTCAAATCGCGGGGCTCAACCCCGTTCCGC
## TACGGAGGATGCGAGCGTTATCCGGATTTATTGGGTTTAAAGGGTGCGTAGGCGGGCTGTTAAGTCAGCGGTCAAATGTCGGGGCTCAACCCCGGCCTGC
## TACGGAGGATGCGAGCGTTATCCGGATTTATTGGGTTTAAAGGGTGCGTAGGCGGGCTTTTAAGTCAGCGGTAAAAATTCGGGGCTCAACCCCGTCCGGC
## TACGGAGGATCCGAGCGTTATCCGGATTTATTGGGTTTAAAGGGAGCGTAGGTGGATTGTTAAGTCAGTTGTGAAAGTTTGCGGCTCAACCGTAAAATTG
## TACGGAGGATGCGAGCGTTATCCGGATTTATTGGGTTTAAAGGGTGCGTAGGCGGCCTGCCAAGTCAGCGGTAAAATTGCGGGGCTCAACCCCGTACAGC
##
## TACGGAGGATGCGAGCGTTATCCGGATTTATTGGGTTTAAAGGGTGCGCAGGCGGAAGATCAAGTCAGCGGTAAAATTGAGAGGCTCAACCTCTTCGAGC
## TACGGAGGATGCGAGCGTTATCCGGATTTATTGGGTTTAAAGGGTGCGCAGGCGGACTCTCAAGTCAGCGGTCAAATCGCGGGGCTCAACCCCGTTCCGC
## TACGGAGGATGCGAGCGTTATCCGGATTTATTGGGTTTAAAGGGTGCGTAGGCGGGCTGTTAAGTCAGCGGTCAAATGTCGGGGCTCAACCCCGGCCTGC
## TACGGAGGATGCGAGCGTTATCCGGATTTATTGGGTTTAAAGGGTGCGTAGGCGGGCTTTTAAGTCAGCGGTAAAAATTCGGGGCTCAACCCCGTCCGGC
## TACGGAGGATCCGAGCGTTATCCGGATTTATTGGGTTTAAAGGGAGCGTAGGTGGATTGTTAAGTCAGTTGTGAAAGTTTGCGGCTCAACCGTAAAATTG
## TACGGAGGATGCGAGCGTTATCCGGATTTATTGGGTTTAAAGGGTGCGTAGGCGGCCTGCCAAGTCAGCGGTAAAATTGCGGGGCTCAACCCCGTACAGC
```

```
taxa.print <- taxa # Removing sequence rownames for display only
rownames(taxa.print) <- NULL
head(taxa.print)
```

```
##      Kingdom    Phylum         Class          Order           Family
## [1,] "Bacteria" "Bacteroidota" "Bacteroidia" "Bacteroidales" "Muribaculaceae"
## [2,] "Bacteria" "Bacteroidota" "Bacteroidia" "Bacteroidales" "Muribaculaceae"
## [3,] "Bacteria" "Bacteroidota" "Bacteroidia" "Bacteroidales" "Muribaculaceae"
## [4,] "Bacteria" "Bacteroidota" "Bacteroidia" "Bacteroidales" "Muribaculaceae"
## [5,] "Bacteria" "Bacteroidota" "Bacteroidia" "Bacteroidales" "Bacteroidaceae"
## [6,] "Bacteria" "Bacteroidota" "Bacteroidia" "Bacteroidales" "Muribaculaceae"
##      Genus
## [1,] NA
## [2,] NA
## [3,] NA
## [4,] NA
## [5,] "Bacteroides"
```

```
## [6,] NA
```

```
library(DECIPHER); packageVersion("DECIPHER")
```

```
## Loading required package: Biostrings

## Loading required package: BiocGenerics

## Loading required package: generics

##
## Attaching package: 'generics'

## The following objects are masked from 'package:base':
##
##     as.difftime, as.factor, as.ordered, intersect, is.element, setdiff,
##     setequal, union

##
## Attaching package: 'BiocGenerics'

## The following objects are masked from 'package:stats':
##
##     IQR, mad, sd, var, xtabs

## The following objects are masked from 'package:base':
##
##     anyDuplicated, aperm, append, as.data.frame, basename, cbind,
##     colnames, dirname, do.call, duplicated, eval, evalq, Filter, Find,
##     get, grep, grepl, is.unsorted, lapply, Map, mapply, match, mget,
##     order, paste, pmax, pmax.int, pmin, pmin.int, Position, rank,
##     rbind, Reduce, rownames, sapply, saveRDS, table, tapply, unique,
##     unsplit, which.max, which.min

## Loading required package: S4Vectors

## Loading required package: stats4

##
## Attaching package: 'S4Vectors'

## The following object is masked from 'package:utils':
##
##     findMatches

## The following objects are masked from 'package:base':
##
##     expand.grid, I, unname

## Loading required package: IRanges
```

```
## Loading required package: XVector
```

```
## Loading required package: Seqinfo
```

```
##
## Attaching package: 'Biostrings'
```

```
## The following object is masked from 'package:base':
##
##     strsplit
```

```
## [1] '3.8.1'
```

```
unqs.mock <- seqtab.nochim["Mock",]
unqs.mock <- sort(unqs.mock[unqs.mock>0], decreasing=TRUE) # Drop ASVs absent in the Mock
cat("DADA2 inferred", length(unqs.mock), "sample sequences present in the Mock community.\n")
```

```
## DADA2 inferred 20 sample sequences present in the Mock community.
```

```
mock.ref <- getSequences(file.path(path, "HMP_MOCK.v35.fasta"))
match.ref <- sum(sapply(names(unqs.mock), function(x) any(grepl(x, mock.ref))))
cat("Of those,", sum(match.ref), "were exact matches to the expected reference sequences.\n")
```

```
## Of those, 20 were exact matches to the expected reference sequences.
```

#import into phyloseq

```
library(phyloseq); packageVersion("phyloseq")
```

```
##
## Attaching package: 'phyloseq'
```

```
## The following object is masked from 'package:IRanges':
##
##     distance
```

```
## [1] '1.56.0'
```

```
library(Biostrings); packageVersion("Biostrings")
```

```
## [1] '2.80.2'
```

```
library(ggplot2); packageVersion("ggplot2")
```

```
## [1] '4.0.3'
```

```r
theme_set(theme_bw())
```

```r
samples.out <- rownames(seqtab.nochim)
subject <- sapply(strsplit(samples.out, "D"), `[`, 1)
gender <- substr(subject,1,1)
subject <- substr(subject,2,999)
day <- as.integer(sapply(strsplit(samples.out, "D"), `[`, 2))
samdf <- data.frame(Subject=subject, Gender=gender, Day=day)
samdf$When <- "Early"
samdf$When[samdf$Day>100] <- "Late"
rownames(samdf) <- samples.out
```

```r
ps <- phyloseq(otu_table(seqtab.nochim, taxa_are_rows=FALSE),
               sample_data(samdf),
               tax_table(taxa))
ps <- prune_samples(sample_names(ps) != "Mock", ps) # Remove mock sample
```

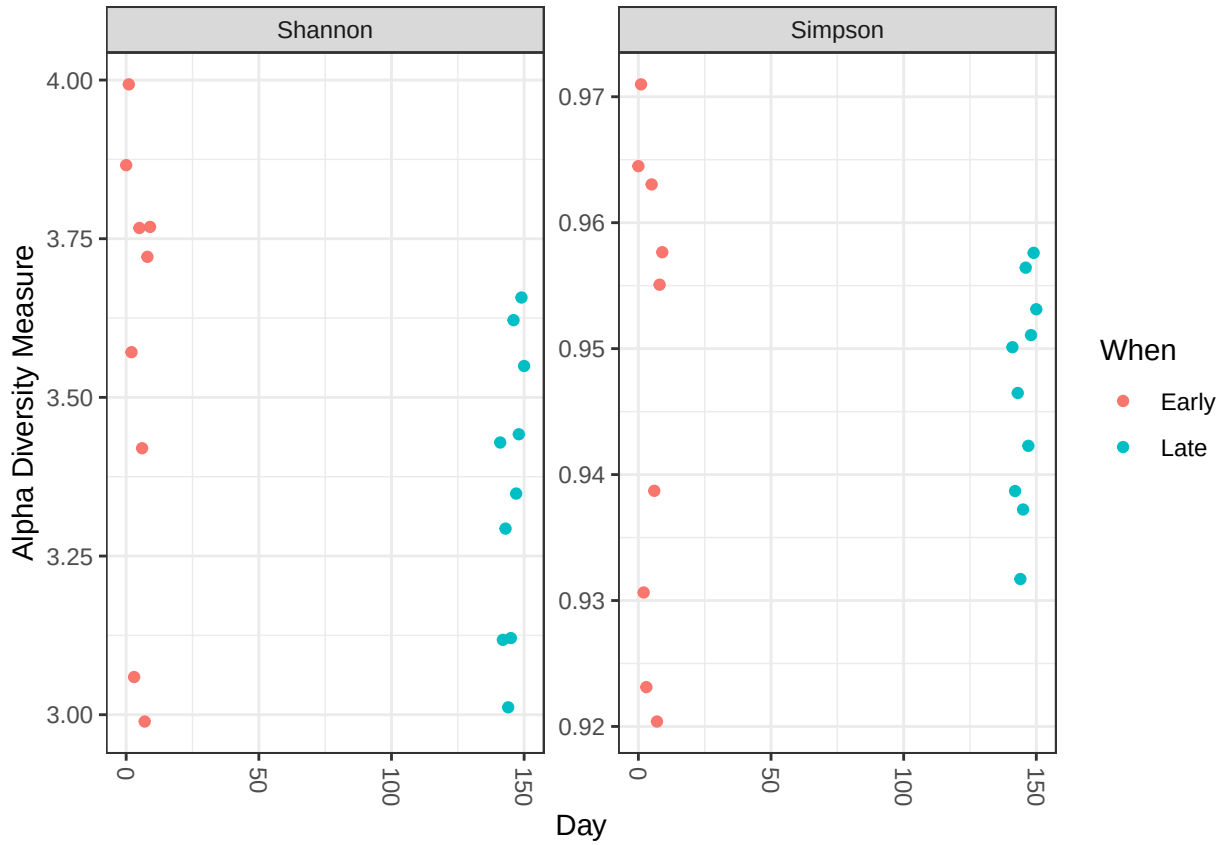
```r
dna <- Biostrings::DNAStringSet(taxa_names(ps))
names(dna) <- taxa_names(ps)
ps <- merge_phyloseq(ps, dna)
taxa_names(ps) <- paste0("ASV", seq(ntaxa(ps)))
ps
```

```
## phyloseq-class experiment-level object
## otu_table()   OTU Table:         [ 232 taxa and 19 samples ]
## sample_data() Sample Data:       [ 19 samples by 4 sample variables ]
## tax_table()   Taxonomy Table:    [ 232 taxa by 6 taxonomic ranks ]
## refseq()      DNAStringSet:      [ 232 reference sequences ]
```

```r
plot_richness(ps, x="Day", measures=c("Shannon", "Simpson"), color="When")
```

```
## Warning in estimate_richness(physeq, split = TRUE, measures = measures): The data you have provided c
## any singletons. This is highly suspicious. Results of richness
## estimates (for example) are probably unreliable, or wrong, if you have already
## trimmed low-abundance taxa from the data.
##
## We recommended that you find the un-trimmed data and retry.
```

```
## Warning: ‘aes_string()‘ was deprecated in ggplot2 3.0.0.
## i Please use tidy evaluation idioms with ‘aes()‘.
## i See also ‘vignette("ggplot2-in-packages")‘ for more information.
## i The deprecated feature was likely used in the phyloseq package.
##   Please report the issue at <https://github.com/joey711/phyloseq/issues>.
## This warning is displayed once per session.
## Call ‘lifecycle::last_lifecycle_warnings()‘ to see where this warning was
## generated.
```

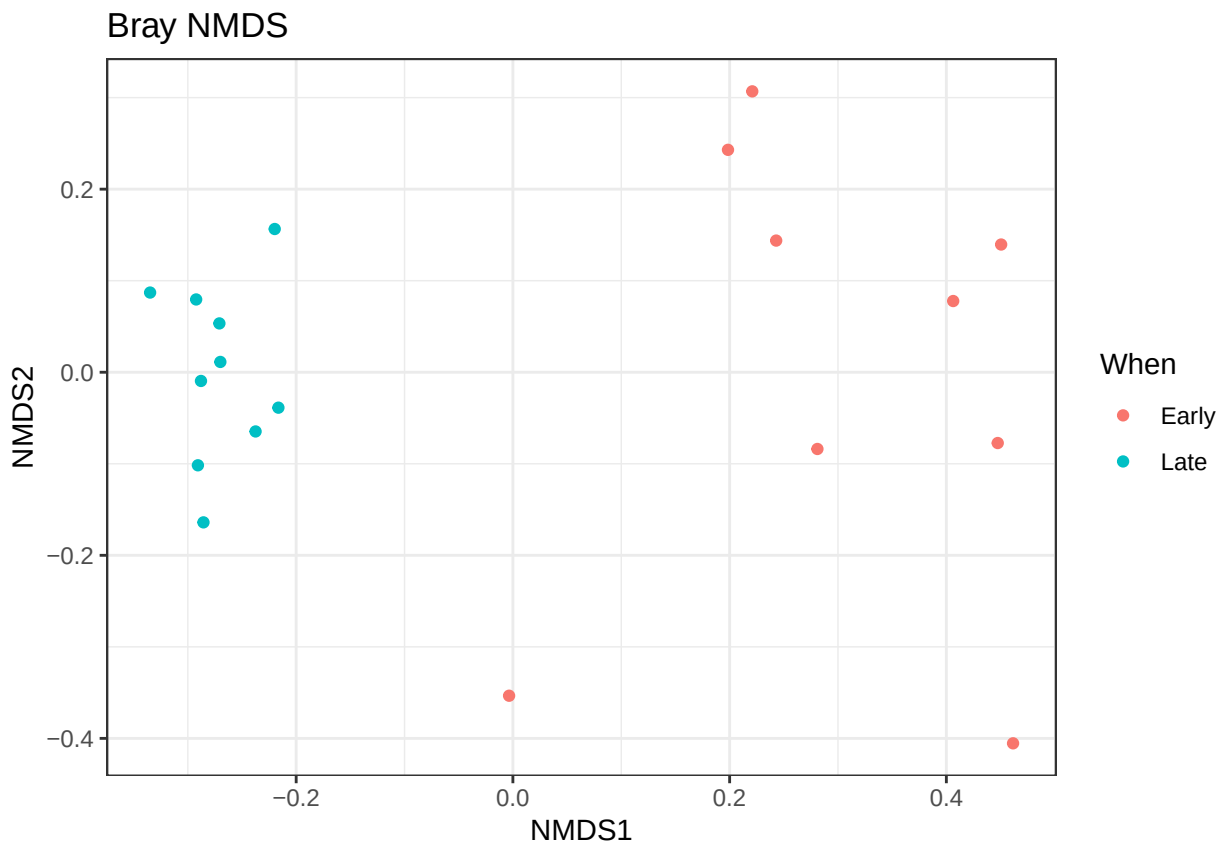



```
# Transform data to proportions as appropriate for Bray-Curtis distances
ps.prop <- transform_sample_counts(ps, function(otu) otu/sum(otu))
ord.nmds.bray <- ordinate(ps.prop, method="NMDS", distance="bray")
```

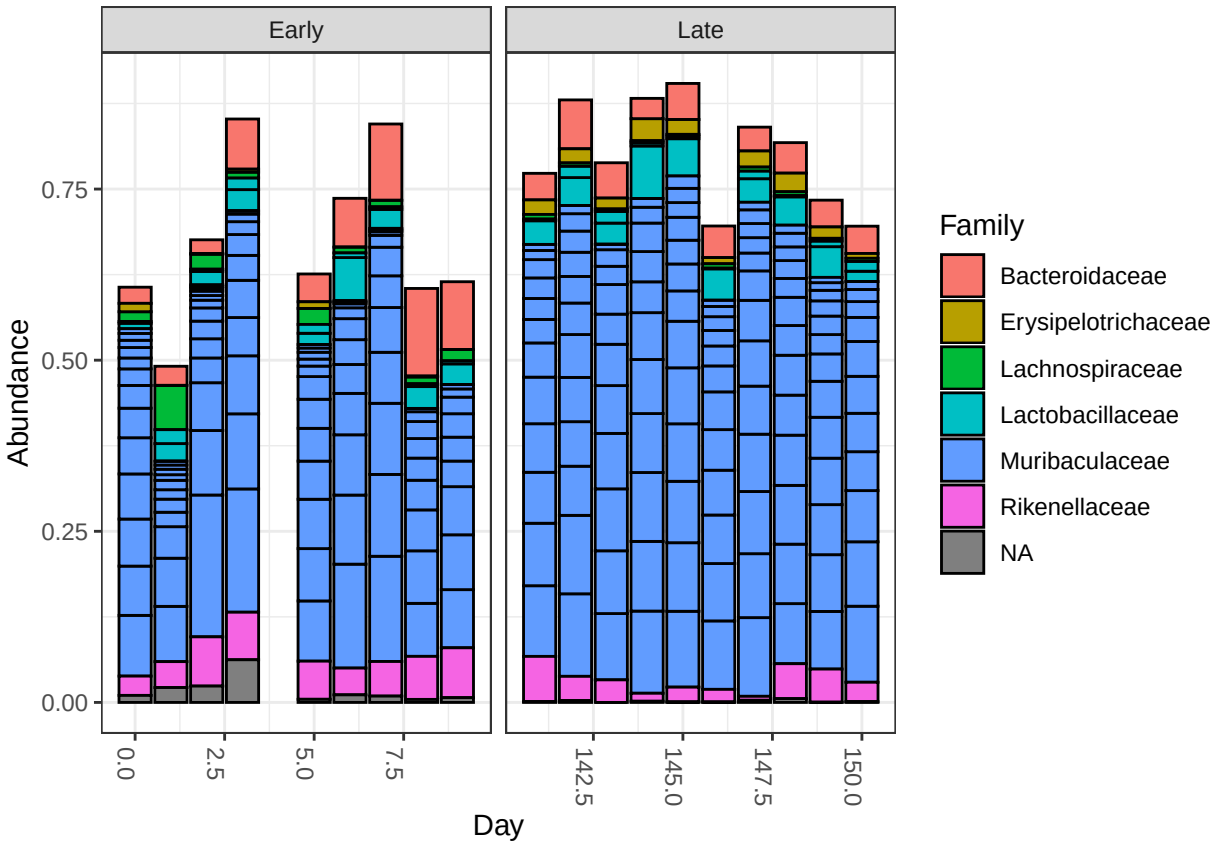
```
## Run 0 stress 0.08043117
## Run 1 stress 0.09477213
## Run 2 stress 0.08076338
## ... Procrustes: rmse 0.0105385  max resid 0.03244034
## Run 3 stress 0.1262108
## Run 4 stress 0.08043117
## ... Procrustes: rmse 4.949841e-06  max resid 1.316944e-05
## ... Similar to previous best
## Run 5 stress 0.09477214
## Run 6 stress 0.1358144
## Run 7 stress 0.08616061
## Run 8 stress 0.08043117
## ... Procrustes: rmse 1.131714e-06  max resid 2.924196e-06
## ... Similar to previous best
## Run 9 stress 0.0807634
## ... Procrustes: rmse 0.01055783  max resid 0.03250349
## Run 10 stress 0.1212044
## Run 11 stress 0.08616061
## Run 12 stress 0.1010632
## Run 13 stress 0.08076337
## ... Procrustes: rmse 0.01050901  max resid 0.03234406
## Run 14 stress 0.08076339
```

```
## ... Procrustes: rmse 0.01050333  max resid 0.03232628
## Run 15 stress 0.08076337
## ... Procrustes: rmse 0.01050984  max resid 0.03234666
## Run 16 stress 0.0807634
## ... Procrustes: rmse 0.01056306  max resid 0.03252033
## Run 17 stress 0.08616061
## Run 18 stress 0.08076338
## ... Procrustes: rmse 0.01052655  max resid 0.03240127
## Run 19 stress 0.08616061
## Run 20 stress 0.08616061
## *** Best solution repeated 2 times
```

```
plot_ordination(ps.prop, ord.nmds.bray, color="When", title="Bray NMDS")
```




```
top20 <- names(sort(taxa_sums(ps), decreasing=TRUE))[1:20]
ps.top20 <- transform_sample_counts(ps, function(OTU) OTU/sum(OTU))
ps.top20 <- prune_taxa(top20, ps.top20)
plot_bar(ps.top20, x="Day", fill="Family") + facet_wrap(~When, scales="free_x")
```


Early
Late
Abundance
0.00
0.25
0.50
0.75
0.0
2.5
5.0
7.5
142.5
145.0
147.5
150.0
Day
Family
Bacteroidaceae
Erysipelotrichaceae
Lachnospiraceae
Lactobacillaceae
Muribaculaceae
Rikenellaceae
NA